DR. TILL HAUSER

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PROFILE

PhD mathematician with a strong background in probability theory and mathematical modelling, as well as practical experience in Python programming and simulation. Experienced in the structured analysis of complex problems, the development of custom software solutions, and the evaluation of mathematical models.

TECHNICAL SKILLS

Programming
Python: object-oriented and modular software development

Modelling and Simulation
Mathematical modelling, Markov chains, Monte Carlo simulations, analysis and visualisation of simulation results

PROFESSIONAL EXPERIENCE

04/2024 – Present **Research, Software Development and Simulation – Materials Science / Mathematics**
German Aerospace Center (DLR), Cologne
Python-based simulations addressing materials science problems and mathematical analysis of the underlying models, particularly in the context of Markov processes. Continued project work on a part-time basis during a two-year research stay in Chile.

10/2024 – 09/2026 **Research and Software Development – Mathematics**
Pontificia Universidad Católica de Chile, Santiago
Research stay within an independently acquired research project funded by the German Research Foundation (DFG) through the Walter Benjamin Programme. Research focused on dynamical systems and ergodic theory. Development of Python software to investigate mathematical problems and support theoretical research. International scientific collaboration.

04/2023 – 03/2024 **Career Transition and Professional Development**
One-month course, "Advanced Python – Blockchain with Python", at Cimdata Bildungsakademie. Independent study of artificial intelligence and Markov chains using university lecture series.

04/2022 – 03/2023 **Research – Mathematics**
Max Planck Institute for Mathematics, Bonn
Independent research in dynamical systems and ergodic theory.

08/2021 – 03/2022 **Research and Teaching – Mathematics**
Friedrich Schiller University Jena
Research and teaching in mathematics; supervision of academic theses and teaching activities.

11/2011 – 12/2011 **Software Development**
Faculty of Mechanical Engineering, TU Dresden
Contributed to the development of a Python application for controlling an Arduino-based measurement system.

02/2010 – 09/2011 **Software Development and Simulation**
Fraunhofer Institute for Process Engineering and Packaging IVV, Dresden
Independently developed a Delphi application for simulating packaging opening processes based on DXF files. Responsibilities included learning the programming language, mathematical modelling, designing the solution approach, developing the user interface, and implementation. Initially developed as a special academic achievement, followed by employment as a student research assistant from 07/2011.

EDUCATION AND CERTIFICATIONS

06/2023 **Certified Associate in Python Programming (PCAP)** **Python Institute**
PCAP-31-0x examination: 93%.

04/2017 – 07/2021 **PhD in Mathematics** **Friedrich Schiller University Jena**
Degree: **summa cum laude**. Dissertation: *Entropy in the Context of Aperiodic Order*. Dean's Doctoral Award (2022). Funded by the German Academic Scholarship Foundation (Studienstiftung des deutschen Volkes). Research areas: ergodic theory, probability theory, topological dynamics, and thermodynamic formalism.

10/2014 – 03/2017 **Master of Science in Mathematics** **TU Dresden / University College Cork**
Final grade: **1.0** (best possible); minor in Physics; Erasmus+ exchange year in Cork, Ireland. Awarded the Ehrenfried Walter von Tschirnhaus Certificate.

10/2011 – 09/2014 **Bachelor's Degree in Mathematics** **TU Dresden**
Final grade: **1.4**

07/2011 **German University Entrance Qualification (Abitur)** **Glückauf-Gymnasium Dippoldiswalde/Altenberg**
Final grade: **1.7**

LANGUAGES

German – Native
English – Fluent
Spanish – Fluent
Russian – Basic knowledge

ADDITIONAL EXPERIENCE

Organisation of an interdisciplinary seminar on mathematics and music; contribution to a documentary awarded the Saxon Media Education Prize in 2016; extensive band experience as a saxophonist (Ska, Reggae); university teaching and supervision of academic theses; international research collaboration.